DEFENCE JOURNEY

NDA Physics Notes

Laws, formulas and applications for the NDA General Science paper

NDA | PHYSICS

Physics forms the biggest part of the NDA General Science questions. The syllabus is at the Class 11 and 12 level, and the questions are mostly conceptual with simple calculations. Learn the definitions, units and formulas below.

1. Units and Measurement

Quantity	SI unit	Quantity	SI unit
Length	metre (m)	Force	newton (N)
Mass	kilogram (kg)	Work, Energy	joule (J)
Time	second (s)	Power	watt (W)
Temperature	kelvin (K)	Pressure	pascal (Pa)
Electric current	ampere (A)	Frequency	hertz (Hz)
Amount of substance	mole (mol)	Charge	coulomb (C)
Luminous intensity	candela (cd)	Resistance	ohm (Ω)

- The 7 SI base quantities are length, mass, time, electric current, temperature, amount of substance and luminous intensity.
- Scalars have only magnitude (distance, speed, mass, energy). Vectors have magnitude and direction (displacement, velocity, force, momentum).
- Dimensional formula: velocity $[LT^{-1}]$, acceleration $[LT^{-2}]$, force $[MLT^{-2}]$, energy $[ML^2T^{-2}]$.

2. Motion and Laws of Motion

- Equations of uniformly accelerated motion: $\mathbf{v = u + at}$, $\mathbf{s = ut + \frac{1}{2}at^2}$, $\mathbf{v^2 = u^2 + 2as}$.
- Free fall: a = g = 9.8 m/s^2 (about 10 m/s^2). A body dropped from height h takes t = √(2h/g).
- Newton's first law: a body stays at rest or in uniform motion unless a force acts on it (law of **inertia**).
- Newton's second law: **F = ma**. Force is the rate of change of momentum.
- Newton's third law: to every action there is an equal and opposite reaction (rocket, recoil of a gun).
- Momentum p = mv. In the absence of external force, total momentum is conserved.
- Circular motion: centripetal acceleration = v^2/r, centripetal force = mv^2/r (directed towards the centre).
- Friction opposes relative motion. Static friction > kinetic friction. Rolling friction is the smallest.

3. Work, Energy and Power

- Work $W = F \cdot d \cdot \cos\theta$. Unit joule. Work is zero if the force is perpendicular to the motion.
- Kinetic energy = $\frac{1}{2}mv^2$. Potential energy = mgh. Power = work/time. 1 horsepower = 746 W.
- Law of conservation of energy: energy can neither be created nor destroyed, only changed from one form to another.
- Work-energy theorem: work done by the net force = change in kinetic energy.
- 1 kWh = 3.6×10^6 J. Efficiency = (useful output/input) × 100%.

4. Gravitation

- Newton's law: $F = G\, m_1 m_2 / r^2$, where $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}.
- g at the surface of the Earth = 9.8 m/s^2. It is maximum at the poles, minimum at the equator, and decreases with height and depth.
- Mass stays the same everywhere. Weight changes (weight on the Moon is about 1/6 of that on Earth).
- Escape velocity from the Earth = 11.2 km/s. Orbital velocity near the Earth ≈ 7.9 km/s.
- Kepler's laws: planets move in ellipses with the Sun at one focus; the line joining a planet to the Sun sweeps equal areas in equal times; $T^2 \propto r^3$.
- A geostationary satellite has a period of 24 hours and orbits at about 36,000 km above the equator.

5. Properties of Matter

- **Archimedes' principle:** a body immersed in a fluid experiences an upward buoyant force equal to the weight of fluid displaced. A body floats if its density is less than that of the liquid.
- **Pascal's law:** pressure applied to a confined fluid is transmitted equally in all directions (hydraulic lift, brakes).
- Pressure in a liquid = $h\rho g$. Atmospheric pressure = 1.013×10^5 Pa (760 mm of mercury).
- **Bernoulli's principle:** where the speed of a fluid is high, the pressure is low (aeroplane lift, spray gun).
- **Surface tension** makes water drops spherical and allows insects to walk on water. Detergent reduces it.
- **Viscosity** is the internal friction of fluids. It decreases as temperature rises for liquids.
- **Hooke's law:** stress is proportional to strain within the elastic limit.

6. Heat and Thermodynamics

- Temperature scales: K = °C + 273. Water freezes at 0°C (273 K) and boils at 100°C (373 K). °F = (9/5)°C + 32.
- Heat $Q = m\, s\, \Delta T$, where s is the specific heat. Water has a very high specific heat (4200 J kg^{-1} K^{-1}), so it is used as a coolant.
- Latent heat of fusion of ice = 336 kJ/kg (80 cal/g). Latent heat of vaporisation of water = 2260 kJ/kg (540 cal/g).
- Modes of heat transfer: **conduction** (solids), **convection** (fluids), **radiation** (needs no medium; Sun's heat reaches us this way).

- Dark, rough surfaces absorb and radiate heat better. Light, shiny surfaces reflect it.
- Water has maximum density at 4°C.
- First law of thermodynamics: $\Delta Q = \Delta U + \Delta W$ (conservation of energy). Second law: heat flows by itself from hot to cold, and no engine is 100% efficient.

7. Waves and Sound

- Wave speed $v = f\lambda$ (frequency × wavelength). Sound is a **longitudinal** wave and needs a medium. Light is a **transverse** wave and does not.
- Speed of sound in air at 20°C is about 343 m/s (330 m/s at 0°C). It is faster in solids than in liquids, and faster in liquids than in gases.
- Audible range for humans: 20 Hz to 20,000 Hz. Below 20 Hz is infrasonic, above 20 kHz is ultrasonic.
- **Echo** is heard if the reflecting surface is at least about 17 m away. **Reverberation** is the persistence of sound due to repeated reflection.
- **Doppler effect:** the apparent change in frequency due to relative motion of the source and listener. It is used in radar and speed guns.
- Pitch depends on frequency, loudness on amplitude, and quality (timbre) on waveform. Loudness is measured in decibels (dB).
- Ultrasound is used in SONAR, medical scanning and cleaning of delicate parts.

8. Light and Optics

- Speed of light in vacuum $c = 3 \times 10^8$ m/s. Light takes about 8 minutes 20 seconds to come from the Sun and 1.3 s from the Moon.
- Laws of reflection: angle of incidence = angle of reflection, and the incident ray, reflected ray and normal lie in one plane.
- Mirror formula: $1/v + 1/u = 1/f$. Lens formula: $1/v - 1/u = 1/f$. Magnification $m = v/u$ (mirror: $-v/u$). Power of a lens $P = 1/f$ (in metres), unit dioptre.
- Concave mirror: used as a shaving mirror, in torches and headlights, and by dentists. Convex mirror: rear-view mirror in vehicles (wider field of view).
- Refractive index $n = c/v = \sin i/\sin r$. Diamond has a very high refractive index (2.42), so it sparkles due to total internal reflection.
- **Total internal reflection** happens when light goes from a denser to a rarer medium at an angle greater than the critical angle. Used in optical fibres and mirages.
- Dispersion: a prism splits white light into VIBGYOR. Violet bends the most and red the least. The rainbow is caused by dispersion and total internal reflection in raindrops.
- The sky looks blue because of scattering of light (blue scatters most). The sun looks red at sunrise and sunset because the longer path scatters away the blue.
- **Eye defects:** Myopia (short-sight) is corrected by a concave lens. Hypermetropia (far-sight) by a convex lens. Presbyopia by bifocal lenses. Astigmatism by cylindrical lenses.

9. Electricity

- **Ohm's law:** $V = IR$. Resistance $R = \rho L/A$. Resistance increases with length and decreases with area.
- Series: $R = R_1 + R_2 + \ldots$ (same current). Parallel: $1/R = 1/R_1 + 1/R_2 + \ldots$ (same voltage).
- Electric power $P = VI = I^2R = V^2/R$. Heat produced $H = I^2Rt$ (Joule's law).
- Fuse wire has a low melting point and is connected in series with the live wire. The earth wire is for safety.
- Domestic supply in India is 220 V, 50 Hz AC.
- Coulomb's law: $F = k\, q_1 q_2 / r^2$. Like charges repel and unlike charges attract.
- Conductors: metals, graphite. Insulators: rubber, glass, plastic. Semiconductors: silicon, germanium.
- A voltmeter is connected in parallel and an ammeter in series.

10. Magnetism and Electromagnetic Induction

- Magnetic field lines go from the North to the South pole outside the magnet and never cross each other.
- A current-carrying wire produces a magnetic field (Oersted). Direction by the right-hand thumb rule.
- Force on a current-carrying conductor in a magnetic field: direction by Fleming's **left-hand** rule. This is the principle of the **electric motor**.
- Faraday's law: a changing magnetic flux induces an emf. Direction by Fleming's **right-hand** rule. This is the principle of the **generator** and the **transformer**.
- A transformer works only on AC. A step-up transformer has more turns in the secondary coil. Power is transmitted at high voltage to reduce loss.
- Lenz's law: the induced current opposes the change that produces it.

11. Modern Physics

- Photoelectric effect (Einstein, Nobel Prize 1921) supports the **particle nature** of light. Energy of a photon $E = h\nu$.
- Radioactivity was discovered by Henri Becquerel. Alpha particles are helium nuclei, beta particles are electrons, and gamma rays are electromagnetic waves (no charge, most penetrating).
- **Nuclear fission:** a heavy nucleus splits (atom bomb, nuclear reactors, uranium-235). **Nuclear fusion:** light nuclei join (hydrogen bomb, Sun's energy).
- Einstein's mass-energy relation: $E = mc^2$.
- X-rays were discovered by Röntgen and are used in medical imaging. Laser produces coherent, monochromatic light.
- A semiconductor diode allows current in one direction only (rectifier). A transistor is used as an amplifier or switch.
- Electromagnetic spectrum (increasing frequency): radio waves, microwaves, infrared, visible, ultraviolet, X-rays, gamma rays.

TIP: Make a one-page sheet of formulas and units. In the exam, many Physics questions are answered by remembering a unit, a law's name or an everyday application.